Università degli Studi di Milano

Tesi di laurea magistrale Scienze e Tecnologie: Informatica

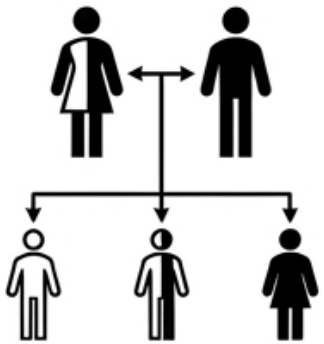
Patient Knowledge Graph Representation Using Non-euclidean Metric

Andrea Moretti

La problematica affrontata: malattie mendeliane

Mutazione monogenetica

Alterazione di un singolo gene



Ereditarietà

Si tramanda di generazione in generazione

Secondo il National Institutes of Health (NIH)

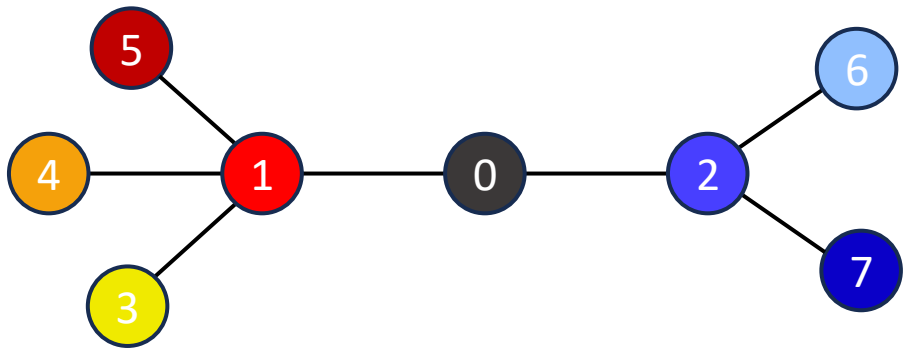
Esistono 10 000 malattie mendeliane distinte

Diagnosticate 2 milioni di persone l'anno

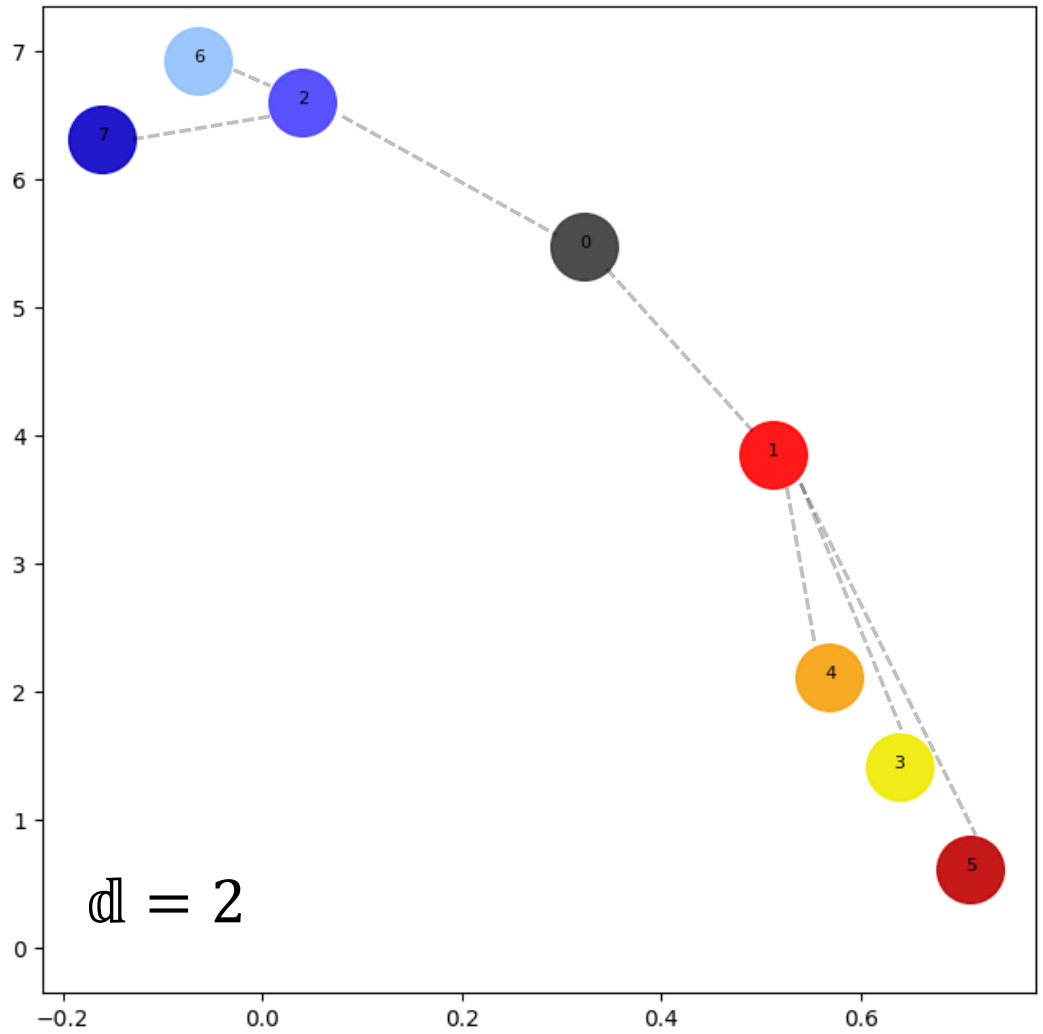
Fonte: American Society of Human Genetics

Oltre il 50% dei pazienti non riceve una diagnosi molecolare certa

Il node embedding



f_{map}



Formalmente: $f_{\text{map}}: V \rightarrow \mathbb{R}^{\mathbb{d}}$

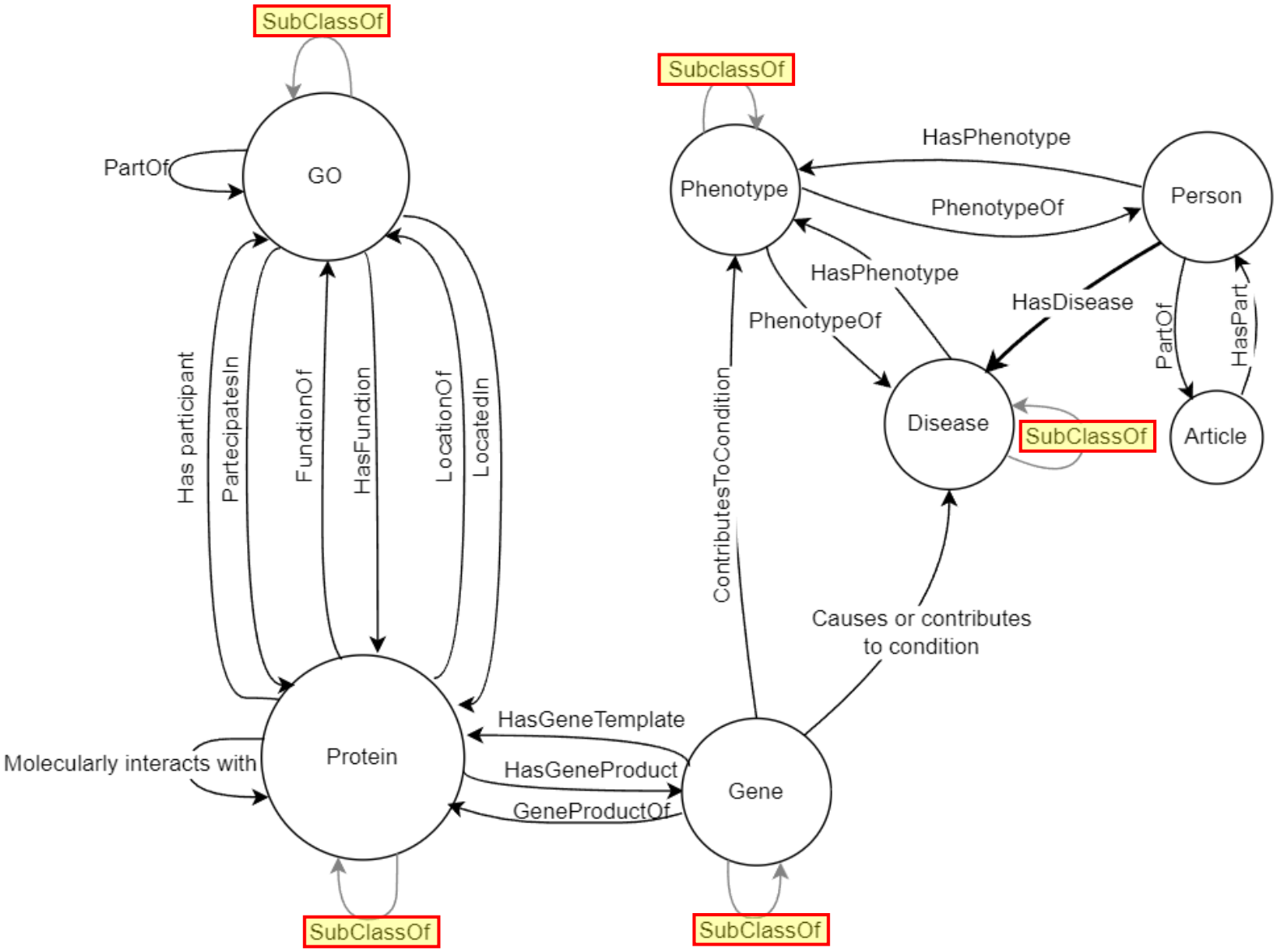
Obiettivi:

- Preservare la topologia
- Ridurre la dimensionalità

Caso studio

Proporzioni del numero di nodi presenti

# Nodi	# Archi
243 053	2 723 540



Caso studio

Proporzioni del numero di nodi presenti

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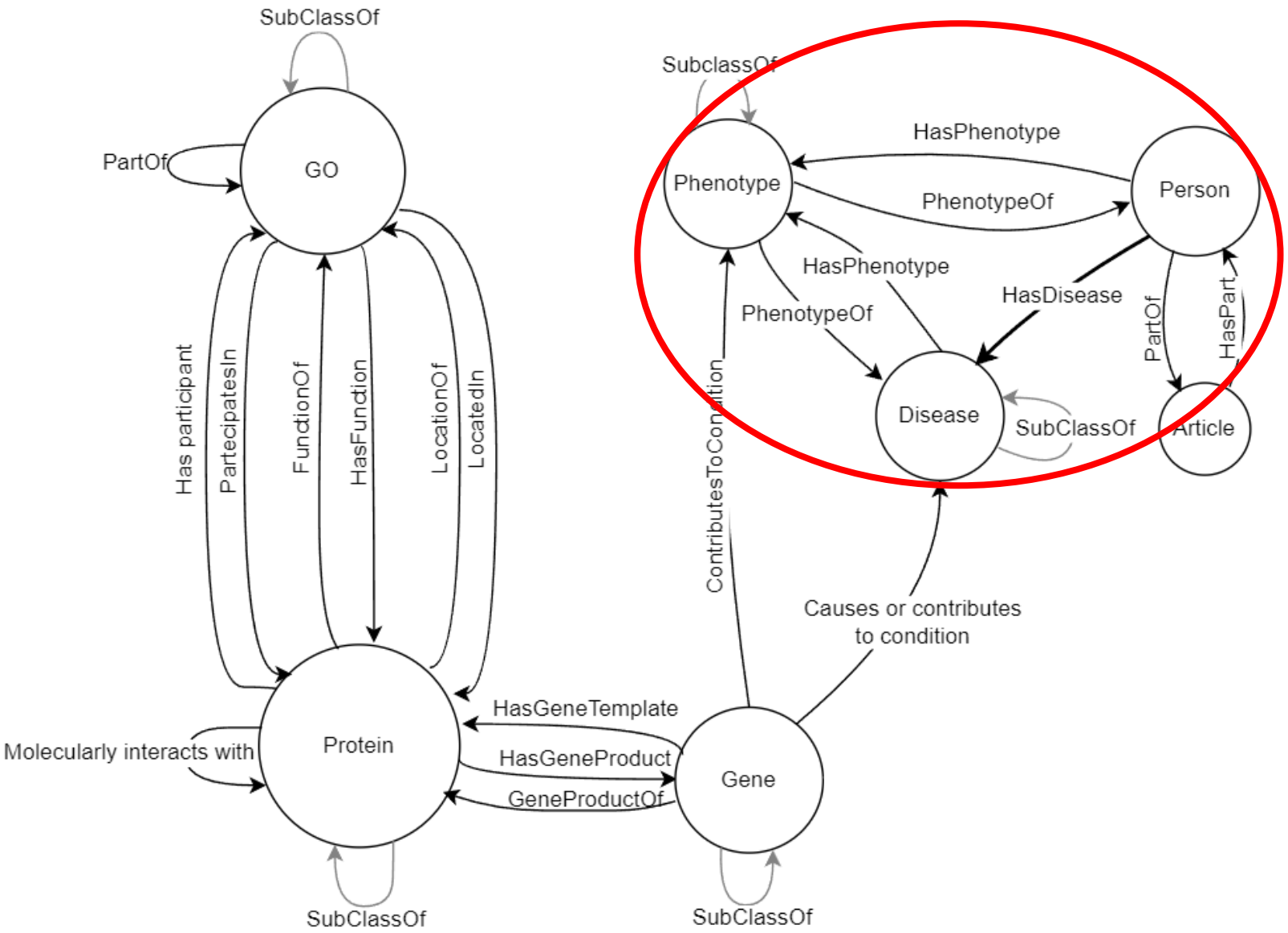


Caso studio

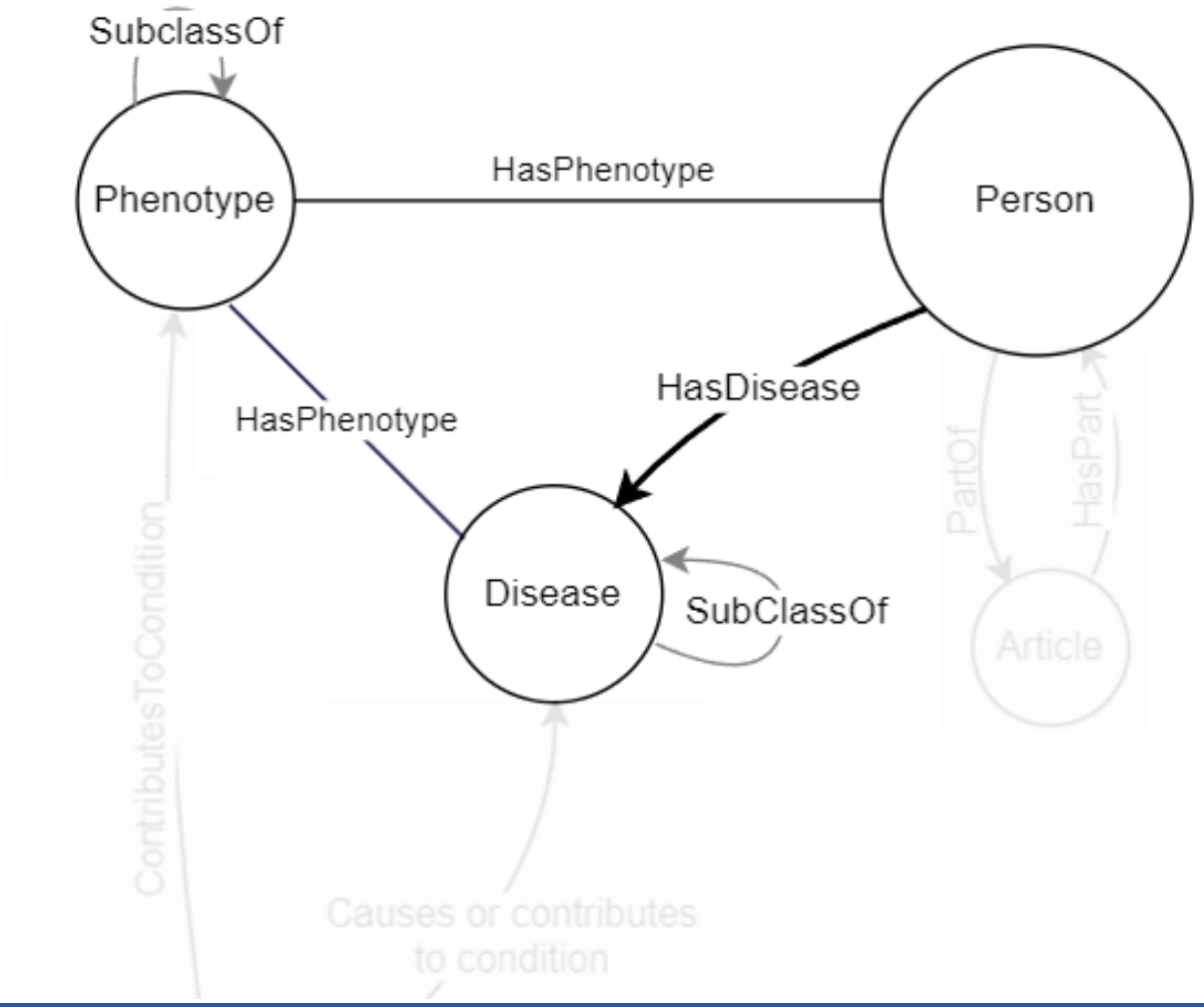
Prima problematica riscontrata:

– Inefficienza della libreria nella gestione della memoria

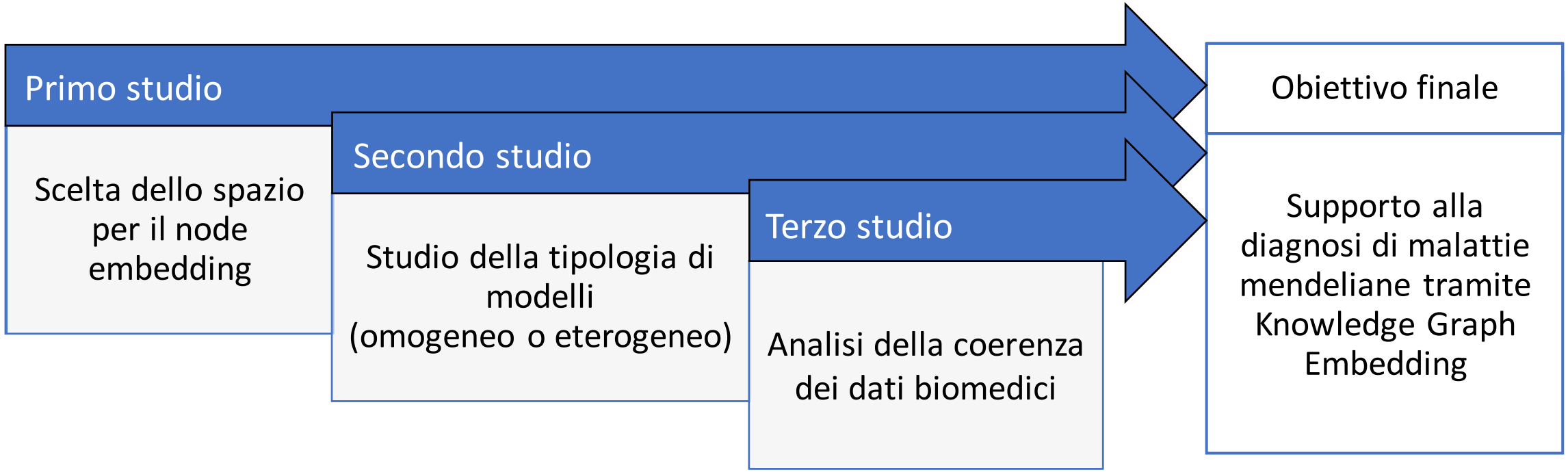
# Nodi	# Archi
243 053	2 723 540

↓ ↓

# Nodi	# Archi
28 811	178 320

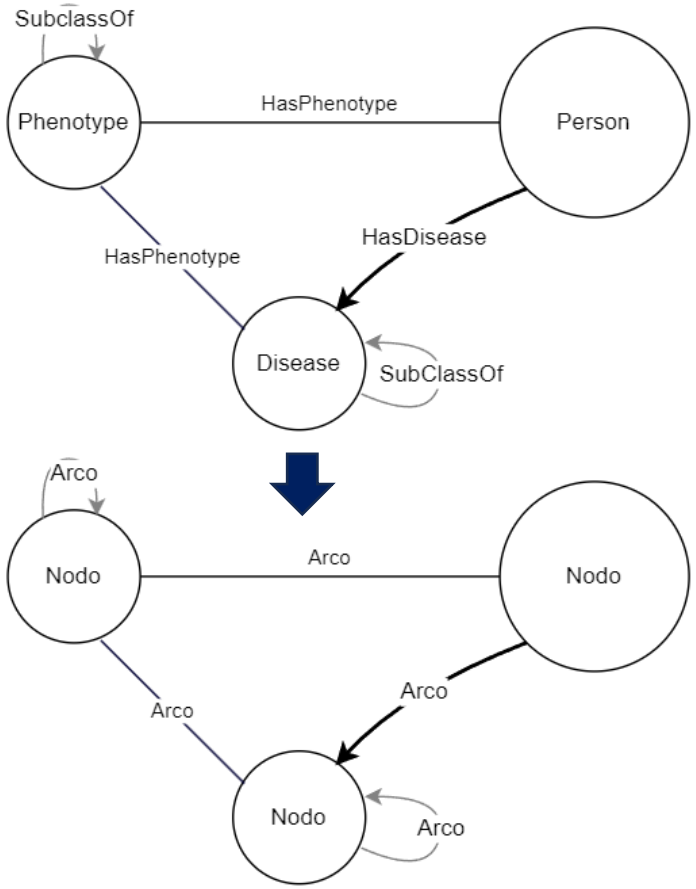


Obiettivi della ricerca

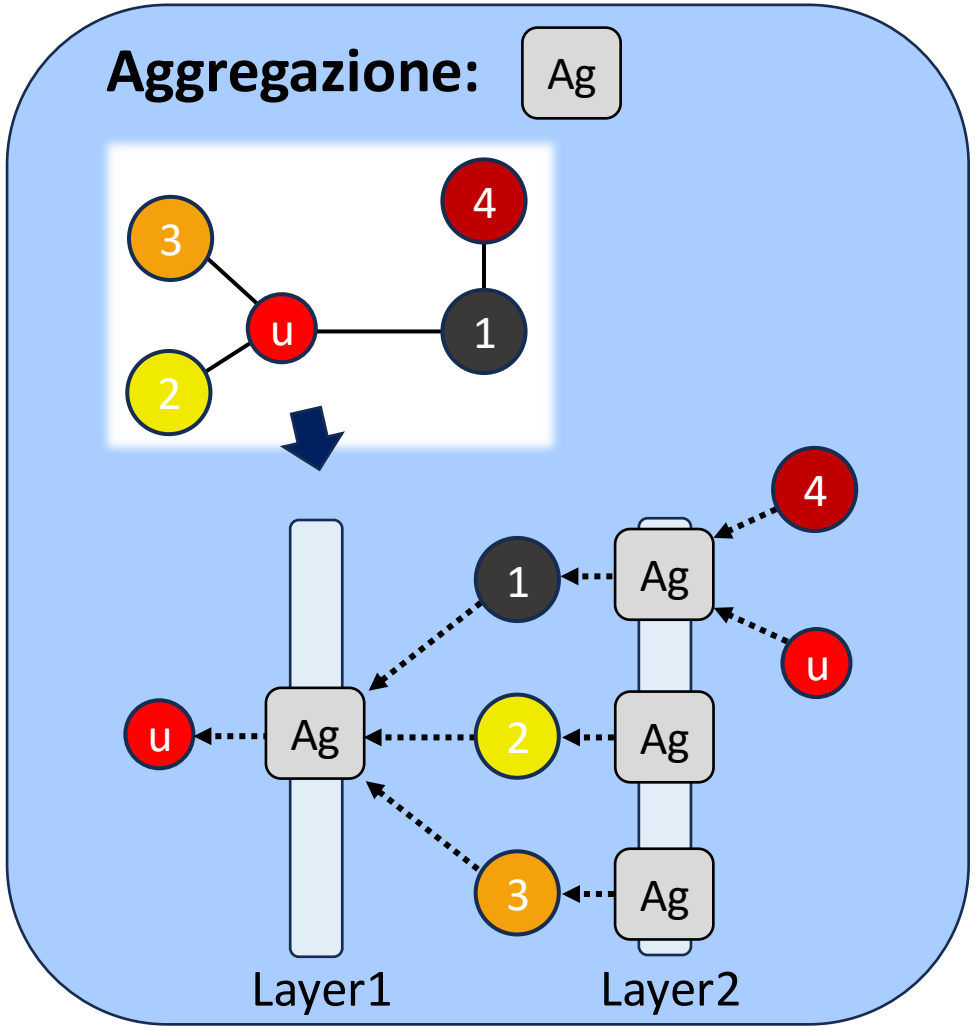


Modelli omogenei: GCN e HGCN

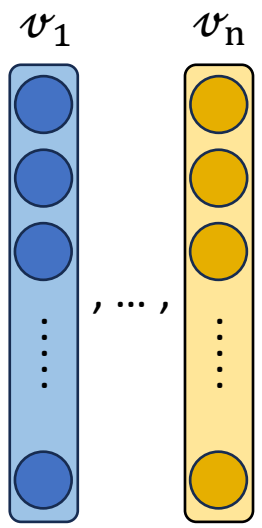
Grafo in input:



Aggregazione:

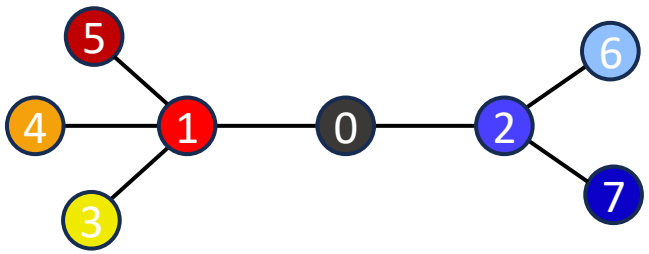


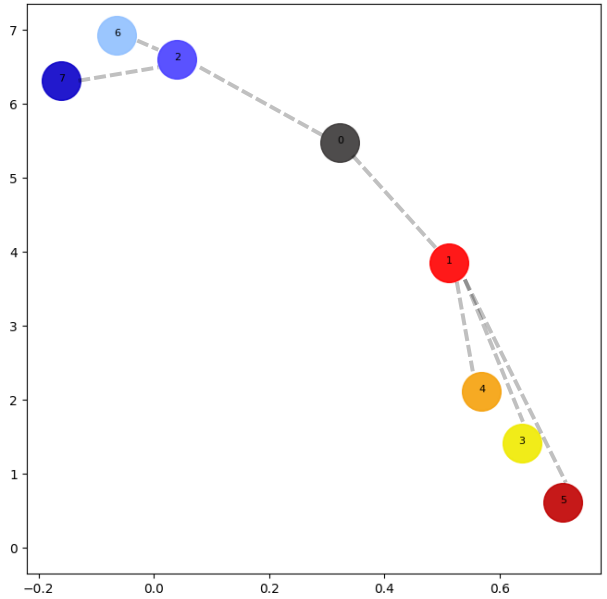
Embedding:



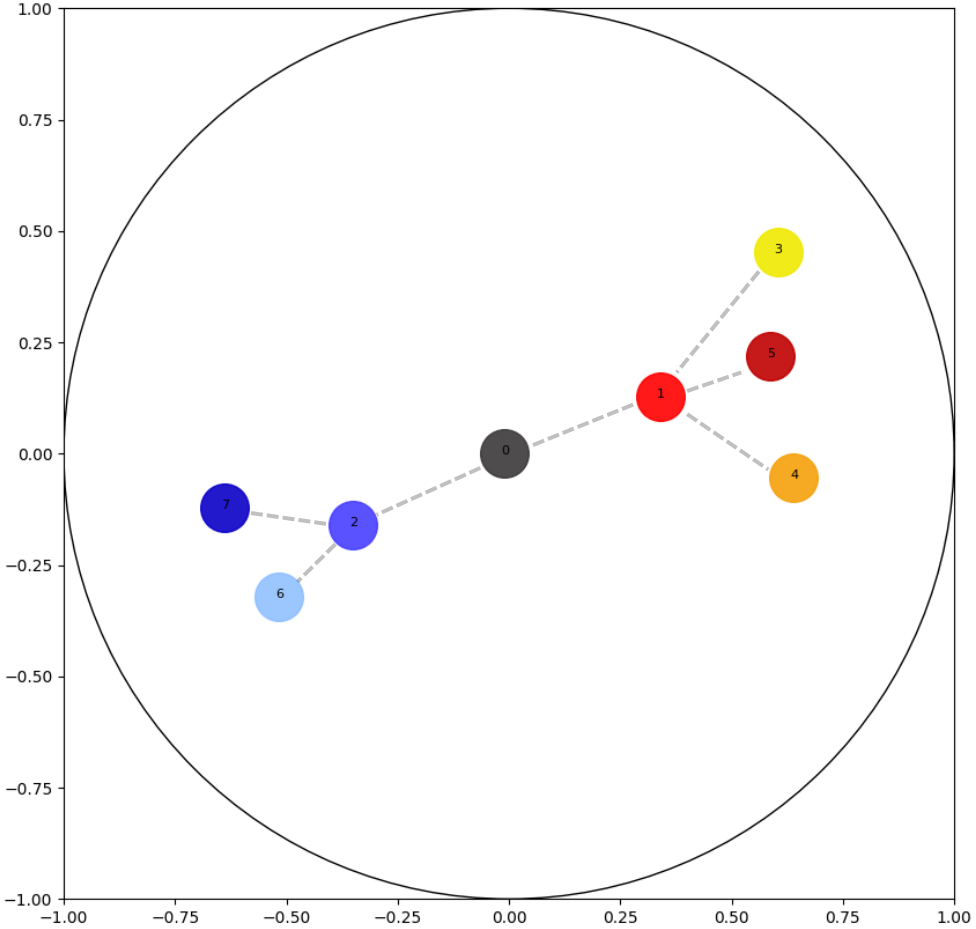
Modelli omogenei: GCN e HGCN

Euclideo:

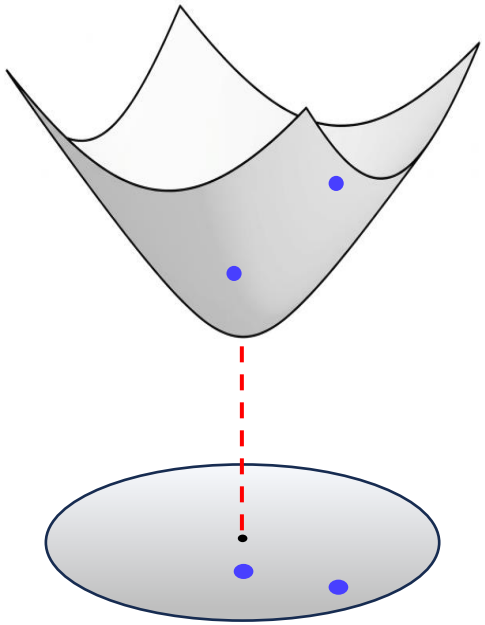




Iperbolico:



Disco di Poincaré



Esempio con $\mathbb{d} = 2$

Configurazione sperimentale

Obiettivo: predire associazione tra paziente e malattia

HasDisease — 7799 record

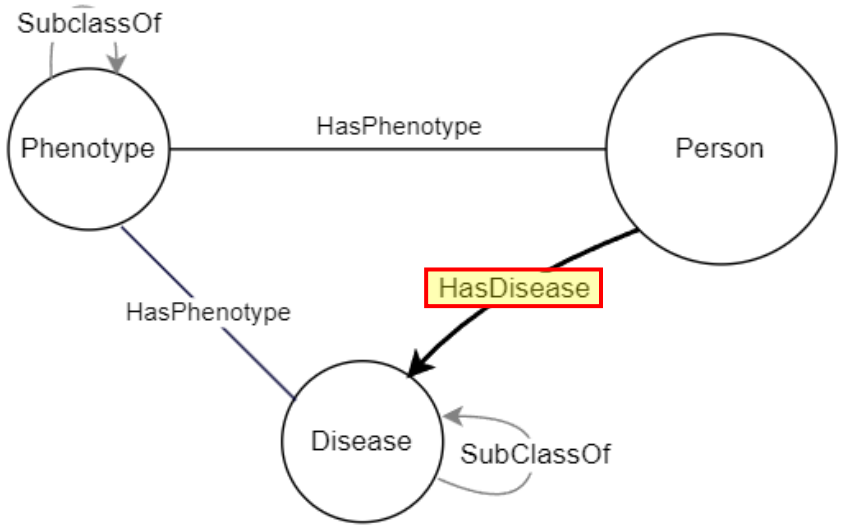
Archi delle 30 malattie più diffuse — 10% degli archi

P — 780 record +

N — 780 record =

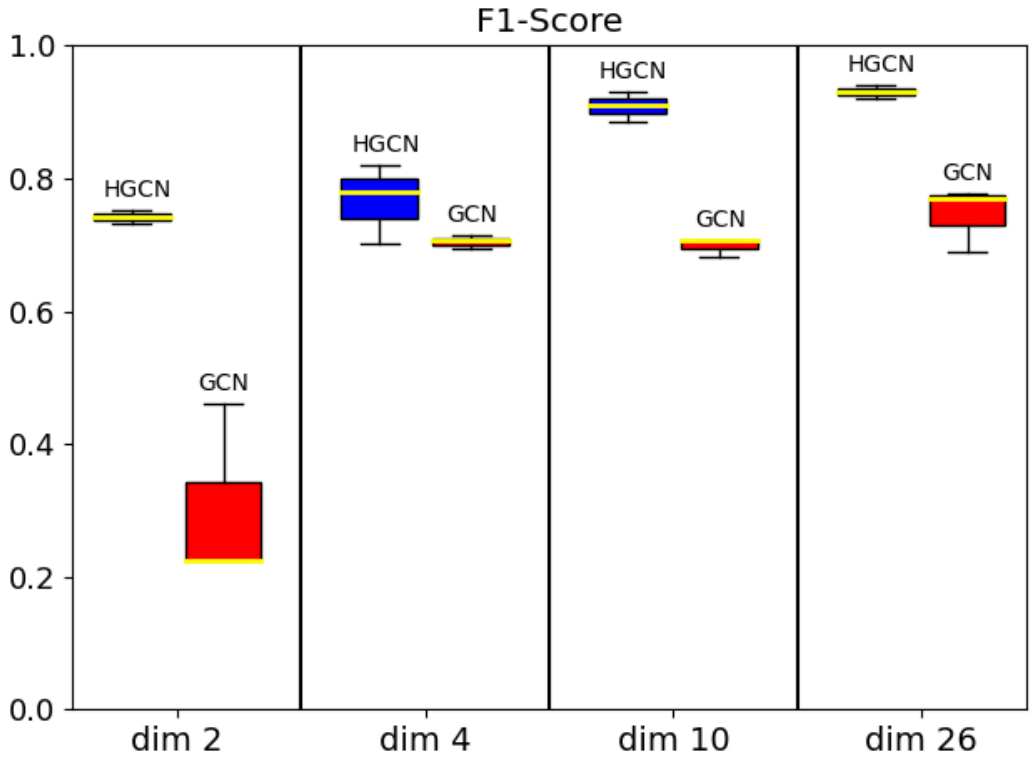
1560 record

Test set bilanciato



Split	Numero di archi	%
Train	168 624	94.5
Validation	8 136	4.5
Test	1 560	1.0
Totale	178 320	100

Applicazione GCN e HGCN al caso studio



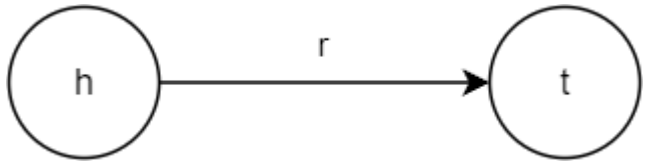
Performance medie su 3 seed differenti

	dim 2		dim 4		dim 10		dim 26	
Metric	GCN	HGCN	GCN	HGCN	GCN	HGCN	GCN	HGCN
ROC-AUC	0.55	**0.74**	0.74	**0.83**	0.67	**0.96**	0.73	**0.97**
Precision	0.24	**0.66**	0.58	**0.76**	0.58	**0.91**	0.62	**0.94**
Recall	0.42	**0.85**	**0.91**	0.79	0.90	**0.91**	**0.94**	0.92
F1-score	0.30	**0.74**	0.70	**0.77**	0.70	**0.91**	0.75	**0.93**

I modelli eterogenei: AttE e AttH

Knowledge Graph Embedding:

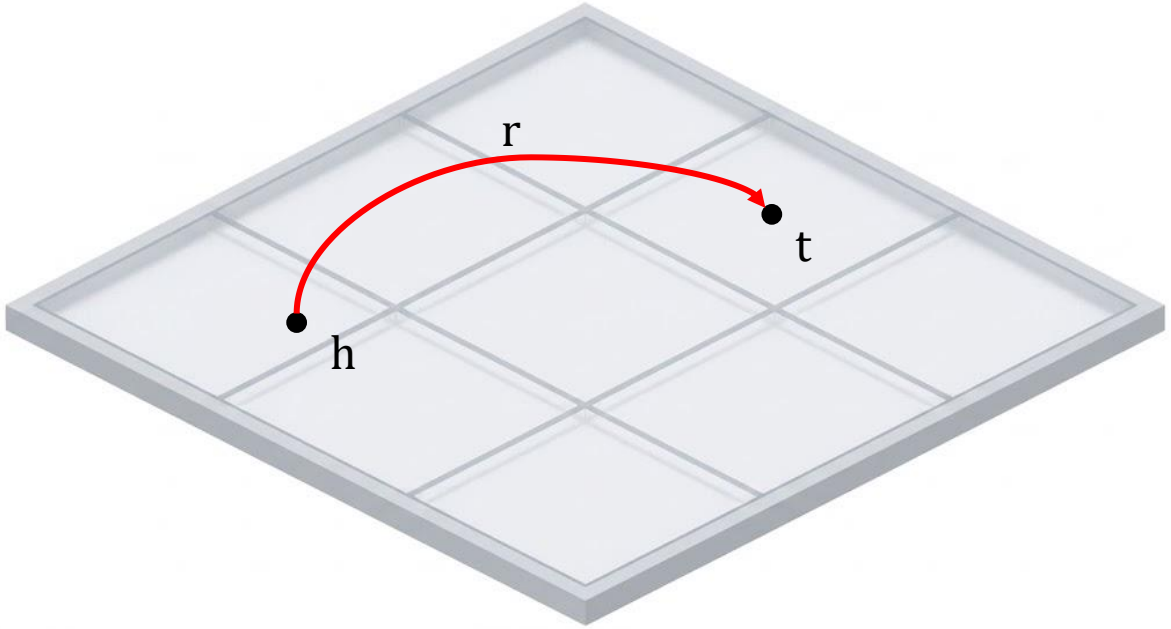
Impara vettori distinti per entità (h, t) e per relazioni (r)



Attention:

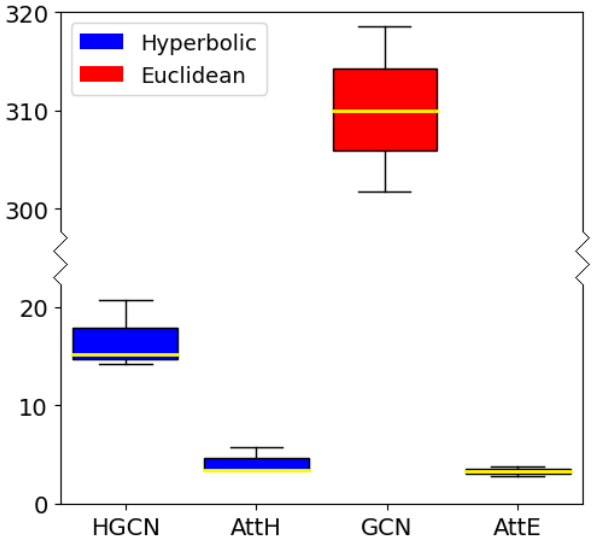
Assegna pesi dinamici per valorizzare le associazioni cliniche più rilevanti

Formalmente: $h \circ r \approx t$

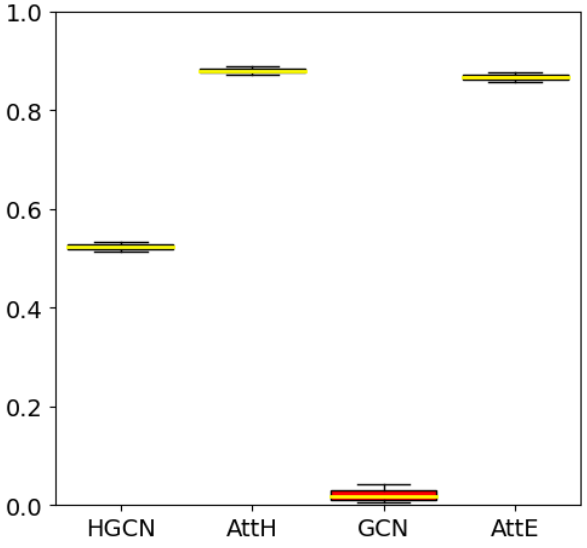


Valutazione efficienza delle predizioni

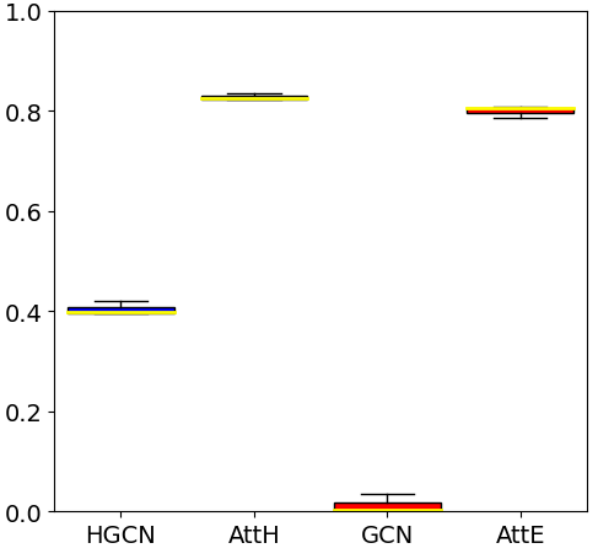
Mean rank (MR) ↓



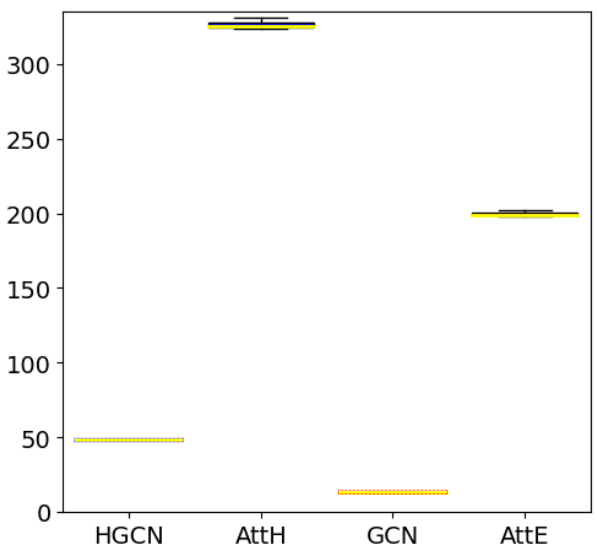
Mean reciprocal rank (MRR)



Hits@1

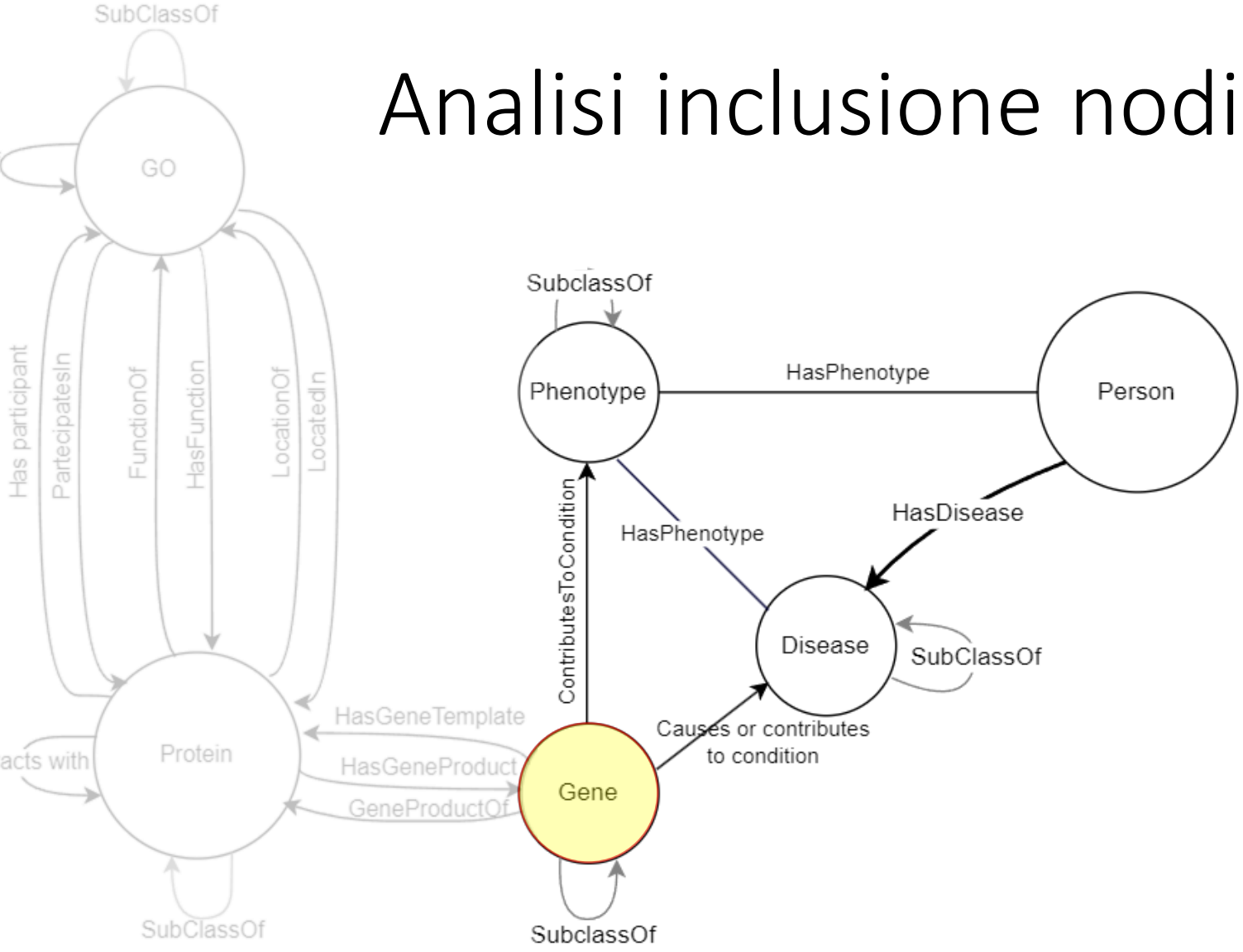


Tempo totale (sec)



Fissando la dimensione degli embedding a 26

Analisi inclusione nodi di tipo Gene



# Nodi	# Archi
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↓ ↓

# Nodi	# Archi
36 171	208 338

Split	Numero di archi	%
Train	197 142	94.6
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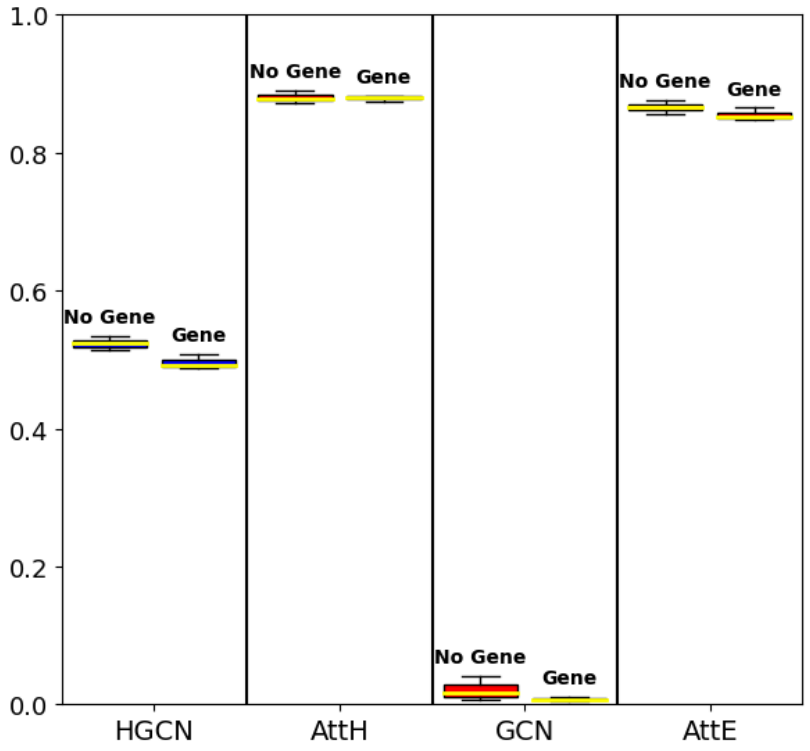
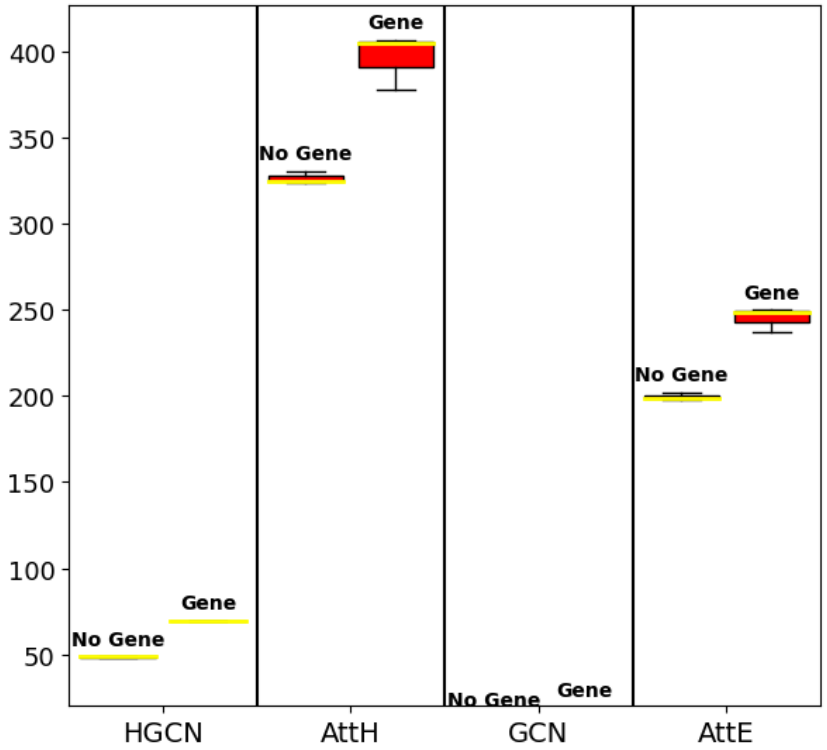
Test	1 560	0.8
Totale	208 338	100

Analisi inclusione nodi di tipo Gene

Mean reciprocal rank (MRR)



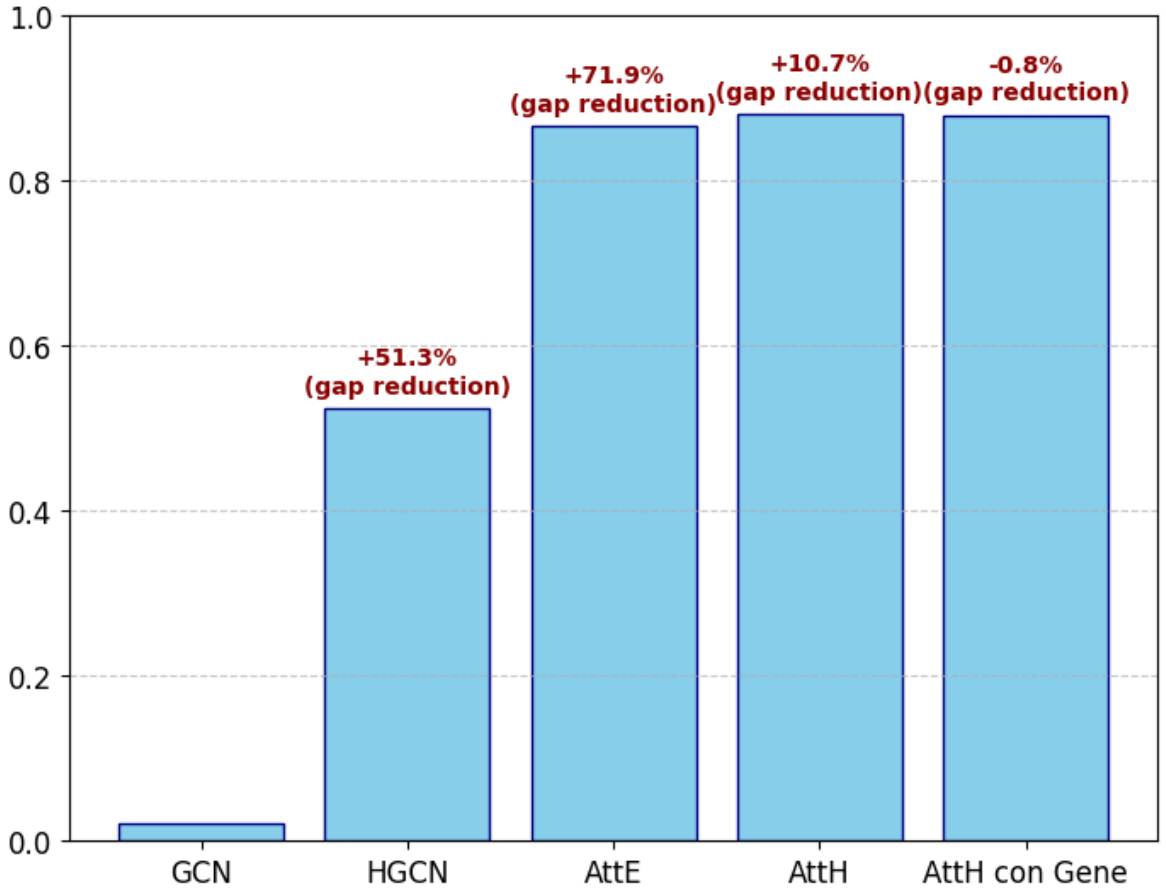
Tempo totale (sec)



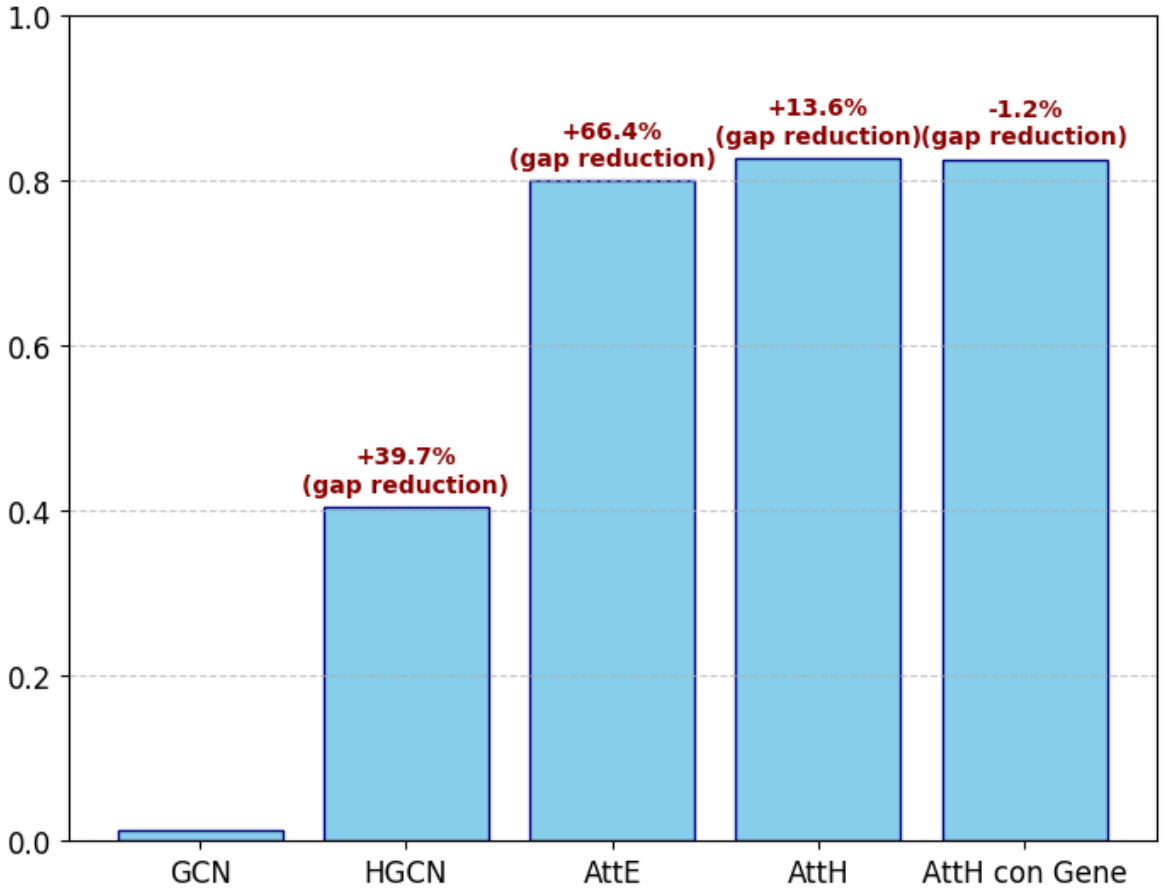
Fissando la dimensione degli embedding a 26

Confronto finale

Mean reciprocal rank (MRR)



Hits@1



Grazie per l’attenzione

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